

Citlalli Trejo Del Rio

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Full-stack developer with experience building scalable web applications, iOS apps, and machine learning systems. Proficient in Java, TypeScript, Swift, and Python with a focus on clean architecture, security, and end-to-end product development.

Education

University of Texas at Dallas

B.S. Computer Science

05/2026

- Member, Association for Computing Machinery
- Member, UTD Society of Automotive Engineers

Professional Experience

[Pancake Money](#)

06/2025 – Present

Frontend Developer

- Led development of an iOS app integrating a Python backend and Plaid API, enabling secure aggregation and visualization of financial data.
- Designed and implemented UI and financial data visualizations in Swift using modular, object-oriented architecture.

Top Escrow

07/2025 – 08/2025

Website Redesign Contractor

- Redesigned and rebuilt the company website using modern frontend practices, improving usability and visual consistency.
- Collaborated with stakeholders to deliver iterative revisions based on feedback
- Supported deployment by configuring domain and ensuring contact form functionality.

Projects

Bigram Language Model & Autocomplete System

01/2026 – 05/2026

Tech Stack: Java 17, Maven, MySQL, JavaFX, JUnit 5

- Architected and implemented a bigram language model in Java using Kneser-Ney smoothing, supporting probabilistic text generation via beam search and greedy decoding.
- Designed a scalable MySQL persistence layer handling 600k+ relationships, optimizing ingestion with batched writes and idempotent upserts.
- Implemented a clean database abstraction layer (DBInterface) to support autocomplete and sentence generation features.
- Wrote unit tests with JUnit 5 covering training, prediction, and edge cases; documented architecture for cross team integration.

Phobos – Wishlist Web App

02/2026 – 04/2026

Tech Stack: Next.js, React, TypeScript, Node.js, Vercel, Railway, PostgreSQL

- Built a full stack wishlist platform with multi-page routing, collaborative sharing, and a custom dark/light theme system.
- Developed a secure REST API with JWT authentication, bcrypt hashing, email verification, and account lockout protection.
- Designed with production readiness in mind, implementing rate limiting, input validation, and secure API practices.
- Deployed on Vercel and Railway under a custom domain, applying best practices including Helmet and CORS.

Partial Convolution U-Net Image Inpainting

08/2025 – 12/2025

Tech Stack: Python, PyTorch, TorchVision, OpenCV, Gradio

- Implemented a Partial Convolution U-Net (33M parameters) for irregular image inpainting based on published research.
- Achieved PSNR of 28–32 dB and SSIM up to 0.96; evaluated performance using FID metrics.
- Built a full training pipeline with logging and an interactive demo interface.

Miel – Pomodoro Timer

06/2025 – Present

Tech Stack: Swift, SwiftUI, Procreate, Adobe Illustrator

- Developed a macOS menu bar timer with customizable UI components and companion features using object-oriented design.
- Rapidly prototyped and debugged features using AI tools and Git based workflows.

Technical Skills

Languages: C++, SQL, Java, HTML, CSS/Tailwind, Swift, JavaScript, TypeScript, Python

Tools & Technologies: Git, VS Code, Linux, Xcode, Figma, React, Node.js, MongoDB, PyTorch, Sheets